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# A Systematic Comparison of Large and Small Language Models in Standalone, Hybrid & Multi-Agent Architectures


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## 1 Abstract / Summary


Measurement-first platform benchmarking **ten architectures in three families** across **five public benchmarks** under a unified observability contract (accuracy · latency · cost · energy · carbon). We introduce **EATS** (Efficiency-Accuracy Trade-off Score) as an architecture-agnostic metric for comparing quality and efficiency together. Routing reaches **EATS 0.893** on HellaSwag and **0.854** on TruthfulQA; Pure Swarm leads GSM8K at **EATS 0.841**; Speculative achieves **EATS 0.908** on ARC. Novel **Blackboard-style** and **swarm** designs show task-specific strengths on reasoning benchmarks.


## 2 Background / Literature Review

**LLMs** offer stronger reasoning but impose higher latency, cost, and energy. **SLMs** are cheaper and easier to self-host, yet usually trail on broad reasoning tasks [1][2].

Our review of **54 collaborative SLM–LLM studies** shows a persistent gap: **accuracy is usually reported, but cost and energy rarely are** [10].

- **RQ1** **Performance.** How do collaborative systems compare with standalone LLMs on reasoning and classification?
- **RQ2** **Efficiency.** How do hybrid and multi-agent designs change cost, latency, and energy?
- **RQ3** **Unified metric.** Does EATS provide a stable cross-architecture trade-off ranking?
- **RQ4** **Empirical grounding.** Do measured results match the trends from the 54-study SLR?

## 3 Measurement-First Platform

Every architecture consumes the same Query and returns the same Response carrying **accuracy, latency, tokens, cost, energy & $CO_2$** per query. The architecture layer is the *only* interchangeable component, so differences are attributable to orchestration logic rather than model API shape [4].

**Experiment Config**
architecture · benchmark · seed · models

↓

**Control Plane**
FastAPI · Next.js UI · runner · host-lock

↓

**Architecture Layer**
routing · speculative · oracle · ensemble · swarm

↓

**Metrics Proxy + vLLM**
latency · GPU power (NVML) · energy · $CO_2$ · cost

↓

**Scorer → MLflow + Reports**
per-item & aggregate metrics

| TIER | HOST | VRAM | MODELS |
|---|---|---|---|
| SLM | NVIDIA L4 | 24 GB | gemma4, qwen3.5, llama3.2 (3–4B) |
| Mid LLM | NVIDIA RTX PRO6000 | 96 GB | gpt-oss, gemma4, qwen3.5 (20–35B dense + MoE) |
| Heavy LLM | NVIDIA H200 | 141 GB | gpt-oss, llama3.3 (70–120B) |

## 4 EATS · Efficiency–Accuracy Trade-off Score

A single score comparing any architecture against the standalone-LLM baseline.

$$\mathrm{EATS} = \frac{Accuracy}{Accuracy + \lambda \cdot P_{\mathrm{efficiency}} + \beta \cdot P_{\mathrm{accuracy}}}$$

$$P_{\mathrm{efficiency}} = 0.65 \cdot Cost + 0.20 \cdot Latency + 0.15 \cdot Energy$$

$$P_{\mathrm{accuracy}} = 1 - Accuracy$$

$\beta = 0.60$ $\lambda = 0.40$

all terms normalised per standalone-LLM baseline · $\beta = 0.60$, $\lambda = 0.40$ calibrated via grid search on MMLU validation split

## 5 Energy, Carbon & Cost

Per-request measurements use GPU power sampling (NVIDIA NVML) at the inference host [4].

$Cost$ = GPU rate · time

$E = P_{GPU} \cdot \Delta t$

$Latency = \Delta t$ per request

$CO_2 = k \cdot E$

k = 380 $gCO_2$/kWh (EU avg) · E in joules ÷ $3.6 \times 10^6$

## 6 Benchmarks & Protocol

Five benchmarks span the **capability dimensions most cited in our companion SLR**, letting us test whether efficiency gains persist as tasks shift from knowledge retrieval to chain-of-thought math and adversarial fact-checking [5][6][7][8][9].

| BENCHMARK | WHAT IT MEASURES |
|---|---|
| **MMLU** | **Broad knowledge** — 57 academic disciplines |
| **ARC-C** | **Scientific reasoning** — multi-step physical-world science |
| **GSM8K** | **Mathematical reasoning** — grade-school math, free-form |
| **HellaSwag** | **Commonsense inference** — sentence completion |
| **TruthfulQA** | **Adversarial truthfulness** — calibration vs. persuasive falsehoods |

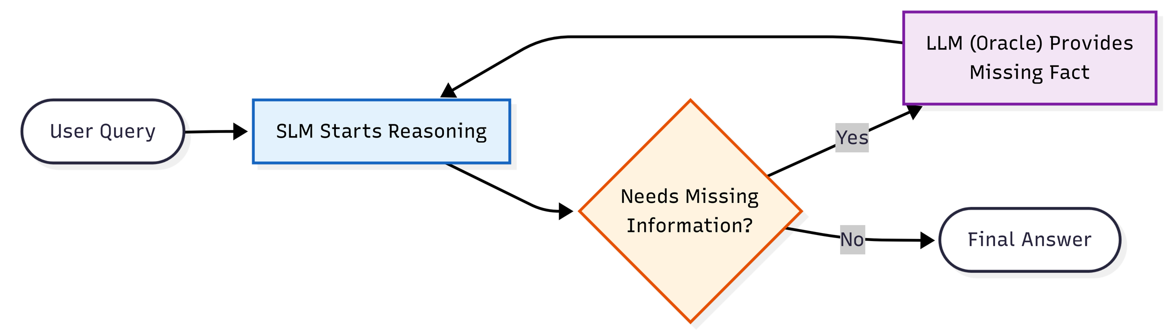
## 7 Architecture Taxonomy

Ten architectures in three families. † marks novel project contributions introduced in the current study.

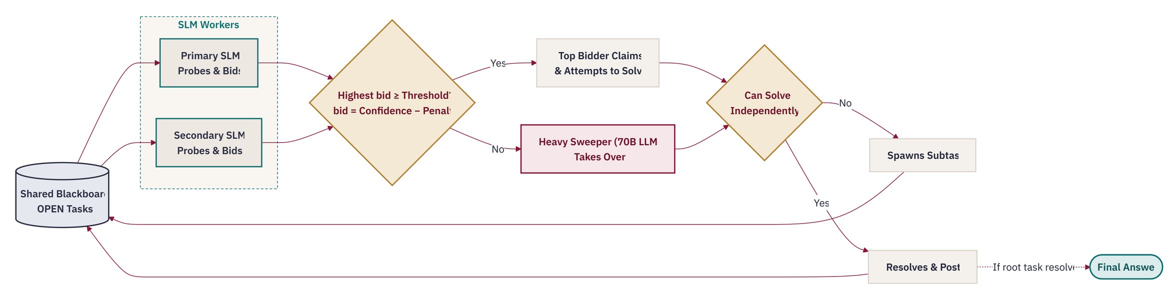
The family progression runs from **internal sparsity** to **explicit coordination** [1][2][3].

| FAMILY | ARCHITECTURE | KEY MECHANISM |
|---|---|---|
| STAND | Single-Model | direct inference |
| STAND | MoE | internal sparse expert activation |
| HYBRID | Routing | confidence SLM→LLM escalation |
| HYBRID | Speculative | SLM drafts; LLM rewrites suffix |
| HYBRID | Debate (POA) | proponent–opponent–arbitrator |
| HYBRID | Active Oracle † | targeted LLM sub-queries |
| HYBRID | Blackboard † | bid-based task board; LLM sweeper fallback |
| HYBRID | Entropy Blackboard † | uncertainty-penalised bidding |
| MULTI | Ensemble | majority or confidence-weighted voting |
| MULTI | Pure Swarm † | SLM-only peer coordination; no LLM online |

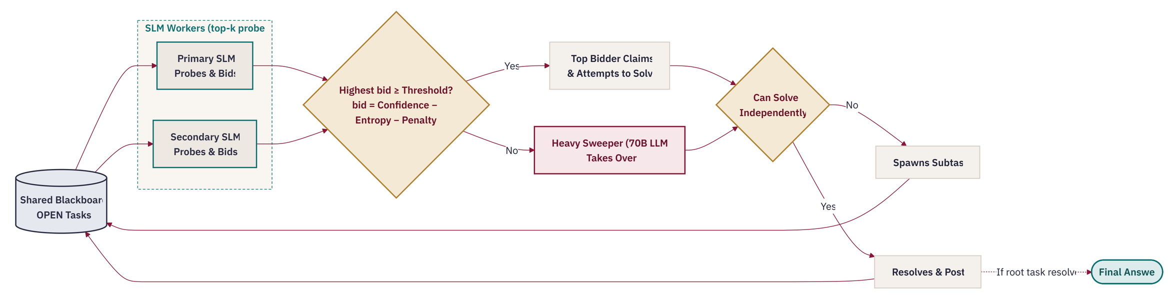
**ACTIVE ORACLE**



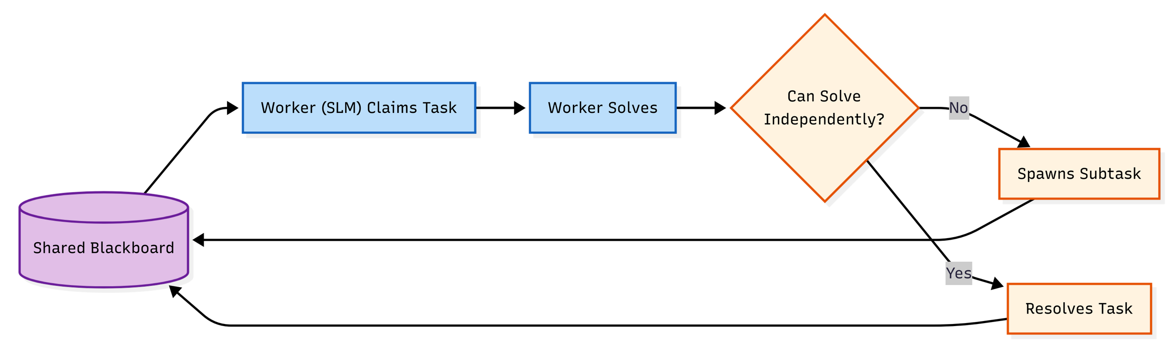

**BLACKBOARD**



**ENTROPY BLACKBOARD**



**PURE SWARM**




## 8 Model Selection & Architecture Pairing

We keep the comparison focused on **architecture behavior** by mostly fixing the core hybrid pair: **Gemma4-4B** as worker and **Gemma4-31B** as verifier/sweeper.

**SELECTION RULE**

Models were chosen for **family diversity**, **open-weight reproducibility**, and **MoE coverage**.

**BEST-WORKING PAIRINGS**

**Routing** and **Speculative** with **Gemma4-4B → Gemma4-31B** gave the strongest EATS on **ARC, HellaSwag, and TruthfulQA**.

- P1 **Routing / Speculative: Gemma4-4B → Gemma4-31B** for the strongest quality-efficiency frontier.
- P2 **Oracle / Blackboard: Gemma4-4B** workers + **Gemma4-31B** sweeper.
- P3 **Standalone MoE: Qwen3.5-35B-A3B** and **Gemma4-26B-A4B** were the strongest efficient baselines.

## 9 UN SDGs

The thesis supports selected **UN Sustainable Development Goals** by making **cost, energy, carbon, and auditability** first-class outputs of architecture evaluation rather than afterthoughts.

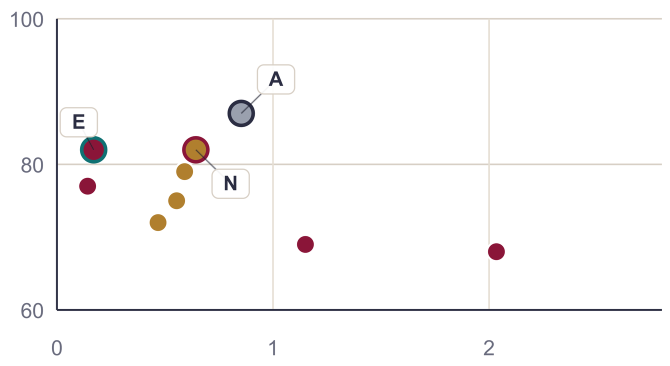
- SDG 8 **Economic efficiency:** cost-aware comparisons show when smaller or hybrid systems are sufficient, reducing unnecessary frontier-model spend.
- SDG 9 **Open infrastructure:** the shared Query/Response contract and common reporting schema create reusable evaluation infrastructure.
- SDG 10 **Reduced inequalities:** hybrid and multi-SLM results show that competitive AI quality can be approached without frontier-scale hardware or cloud budgets.
- SDG 12/13 **Responsible energy use:** EATS includes energy and carbon in the main ranking, so wasteful paths are penalised directly.
- SDG 16 **Accountable evaluation:** versioned benchmarks, host logs, and transparent metrics make architecture claims easier to audit.

The contribution is therefore **infrastructural**: it helps teams choose capable systems with clearer evidence about **quality, cost, and sustainability**.

## 10 Benchmark Readouts

Each mini-plot shows accuracy against efficiency penalty, so the strongest trade-offs appear toward the upper-left. **E** = EATS winner, **A** = accuracy leader, **N** = best novel result.

**MMLU**

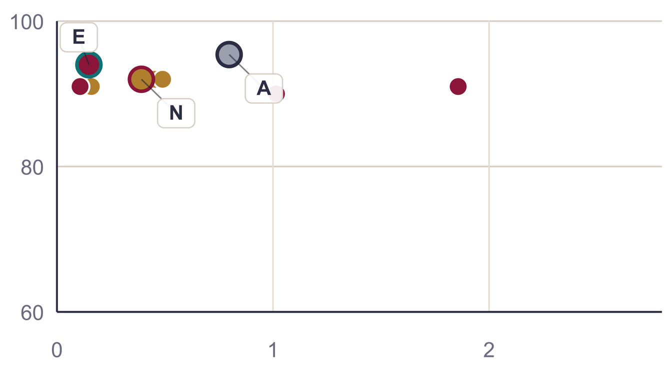



**EATS winner** Speculative

**ACC leader** gpt-oss-120b · 87.0%

**NOVEL best** Entropy BB · 82.0%

**ARC**

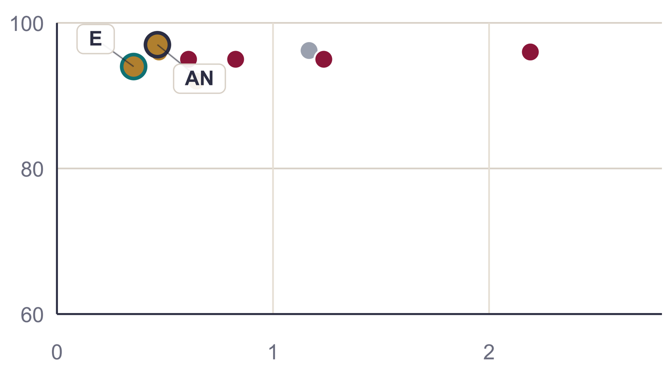



**EATS winner** Speculative

**ACC leader** gpt-oss-120b · 95.4%

**NOVEL best** Pure Swarm · 92.0%

**GSM8K**

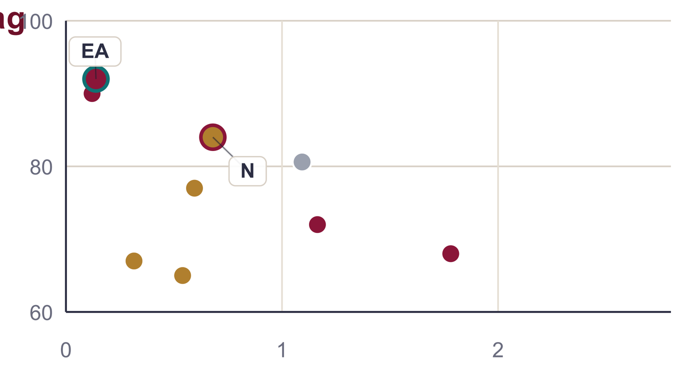



**EATS winner** Pure Swarm

**ACC leader** Entropy BB · 97.0%

**NOVEL best** Entropy BB · 97.0%

**HellaSwag**

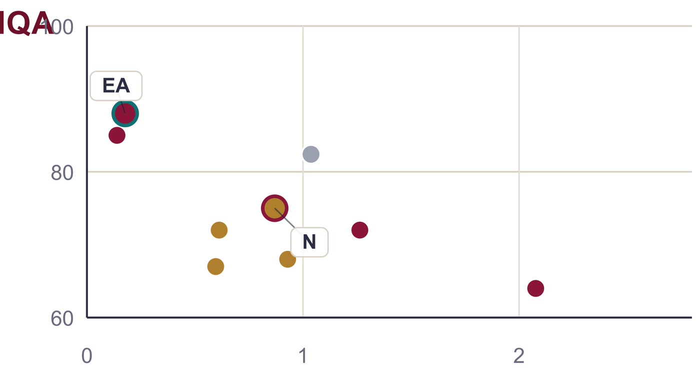



**EATS winner** Speculative

**ACC leader** Speculative · 92.0%

**NOVEL best** Entropy BB · 84.0%

**TruthfulQA**




**EATS winner** Speculative

**ACC leader** Speculative · 88.0%

**NOVEL best** Entropy BB · 75.0%

## 11 EATS Landscape

Canonical EATS for the latest qualified run of each architecture; the standalone row shows the best MoE baseline on each benchmark.

| ARCHITECTURE | MMLU | ARC | GSM8K | HSWAG | TQA |
|---|---|---|---|---|---|
| Routing | 0.80 | 0.90 | 0.78 | 0.89 | 0.85 |
| Speculative | **0.82** | **0.91** | 0.72 | **0.90** | **0.86** |
| Standalone (MoE) | 0.68 | 0.73 | 0.66 | 0.59 | 0.61 |
| Blackboard † | 0.69 | 0.80 | 0.82 | 0.67 | 0.55 |
| Entropy BB † | 0.69 | 0.79 | 0.83 | 0.69 | 0.60 |
| Active O. † | 0.67 | 0.89 | 0.75 | 0.67 | 0.64 |
| Debate POA | 0.52 | 0.66 | 0.64 | 0.53 | 0.52 |
| Pure Swarm † | 0.67 | 0.82 | **0.84** | 0.60 | 0.61 |
| Ensemble | 0.40 | 0.53 | 0.52 | 0.43 | 0.38 |

Darker = higher EATS · bold = benchmark winner · tinted rows / † = novel contribution.

## 12 Conclusions

The main synthesis is not that one model wins everywhere, but that **the best architecture depends on how much capability you can afford to escalate online.** That is why the poster reports both raw accuracy and EATS rather than collapsing the study into a single leaderboard.

- **RQ1** Collaborative systems are competitive with standalone LLMs; the strongest results come from routing, speculative decoding, and novel coordination on reasoning-heavy tasks.
- **RQ2** Hybrid escalation gives the clearest cost and energy gains, while multi-agent coordination usually pays a latency penalty.
- **RQ3** EATS gives a consistent cross-architecture ranking: speculative leads four benchmarks, and Pure Swarm leads GSM8K.
- **RQ4** The measurements corroborate 8 of 10 SLR trends: SLM-first escalation is the strongest general pattern, while coordination and entropy-aware bidding help on task-specific cases.

**LIMITATIONS**

- Single hardware stack (L4 / PRO6000 / H200) — results may not generalise to other GPU tiers or cloud inference endpoints.
- Benchmark scope covers 5 NLP task types; code generation, retrieval-augmented and long-context tasks are not included.
- EATS weights ($\beta = 0.60$, $\lambda = 0.40$) are fixed after calibration; sensitivity to alternative weight choices is not explored in this study.

In the full thesis, these findings will be extended with larger sample sizes, pair ablations, and broader task surfaces, but the poster already shows a stable message: **architecture design can buy back a large fraction of frontier-model quality without always paying frontier-model cost.**


**SELECTED REFERENCES**

[1] Hao et al. **Hybrid SLM–LLM Inference**, 2024 · [2] Oh & Kim, **Uncertainty-Aware Hybrid Inference**, 2025 · [3] Shao et al. **Division-of-Thoughts**, 2025 · [4] Kwon et al. **vLLM: PagedAttention**, SOSP 2023 · [5] Hendrycks et al. **MMLU**, ICLR 2021 · [6] Clark et al. **ARC Challenge Set**, 2018 · [7] Cobbe et al. **GSM8K**, 2021 · [8] Zellers et al. **HellaSwag**, ACL 2019 · [9] Lin et al. **TruthfulQA**, ACL 2022 · [10] Companion SLR (54 studies, PRISMA 2020).



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